ALMA Proposal Preparation Workshop

March 20, 2026 @IFA-UV, Valparaíso

2.3. ALMA Proposal Preparation

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ALMA Proposal Preparation: Overview

CfP deadline: April 23

- **Create an ALMA account**
  https://asa.alma.cl/UserRegistration/newAccount.jsp?arc=eu
  - Permanent username
  - Select expertise keywords
- **Observing Tool (OT)**
  https://cycle-13.sps.alma.cl/ngot/
  - Design the Science Goals in OT, test project feasibility
- **Project preparation**
  - Prepare the science case (SJ; must be strictly anonymous)
  - Justify technical setup (TJ)

Timeline

Key considerations

→ Sensitivity required
→ Angular resolution
→ Spectral resolution
→ Target size scales

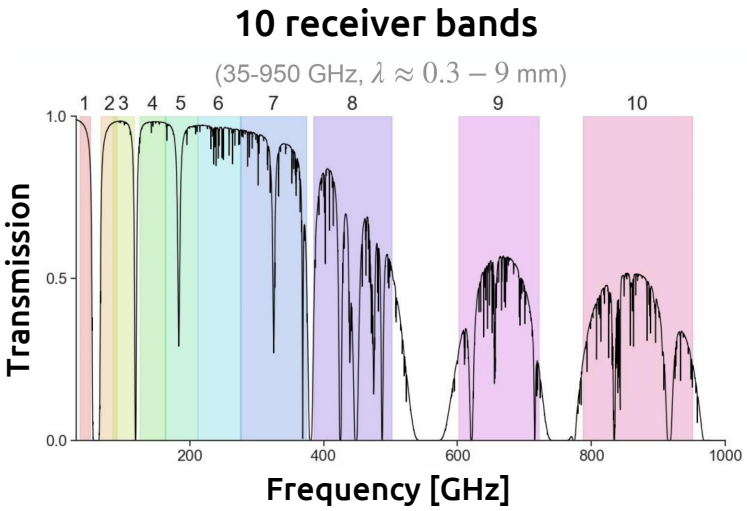
Range of angular scales

$$\lambda/B_{max} < \theta < \lambda/B_{min}$$
$$AR < \theta < LAS$$

AR: Angular Resolution
LAS: Largest Angular Scale

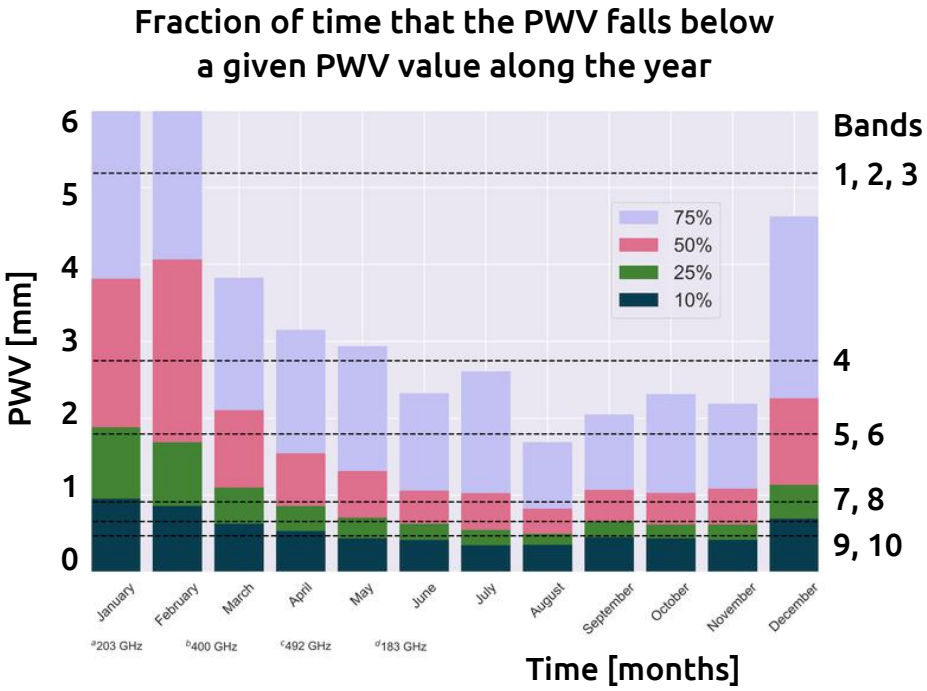
Observing Strategy: Receiver Bands

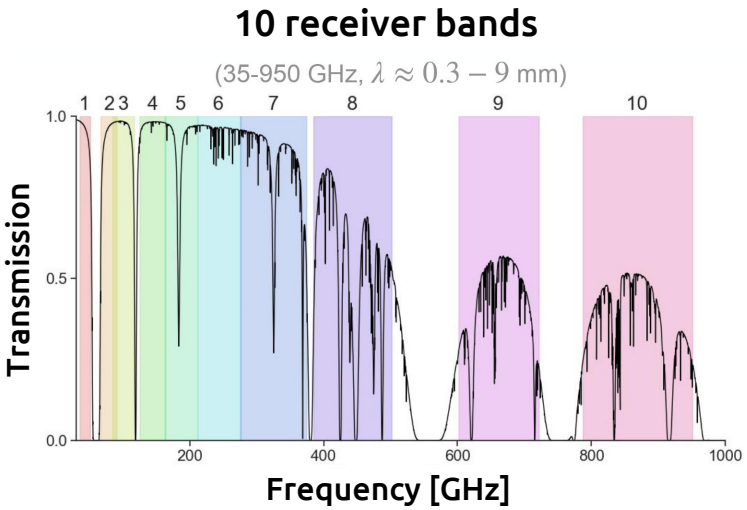
Band	Frequency range [GHz]	Wavelength range [mm]	Type
1	35 – 50	8.5 – 6	SSB
2	67 – 116	4.5 – 2.6	2SB
3	84 – 116	3.6 – 2.6	2SB
4	125 – 163	2.4 – 1.8	2SB
5	158 – 211	1.9 – 1.4	2SB
6	211 – 275	1.4 – 1.1	2SB
7	275 – 373	1.1 – 0.8	2SB
8	385 – 500	0.78 – 0.60	2SB
9	602 – 720	0.50 – 0.42	DSB
10	787 – 950	0.38 – 0.32	DSB



Cycle 13 Proposer's Guide: https://almascience.nrao.edu/proposing/proposers-guide

Observing Strategy: Weather Conditions





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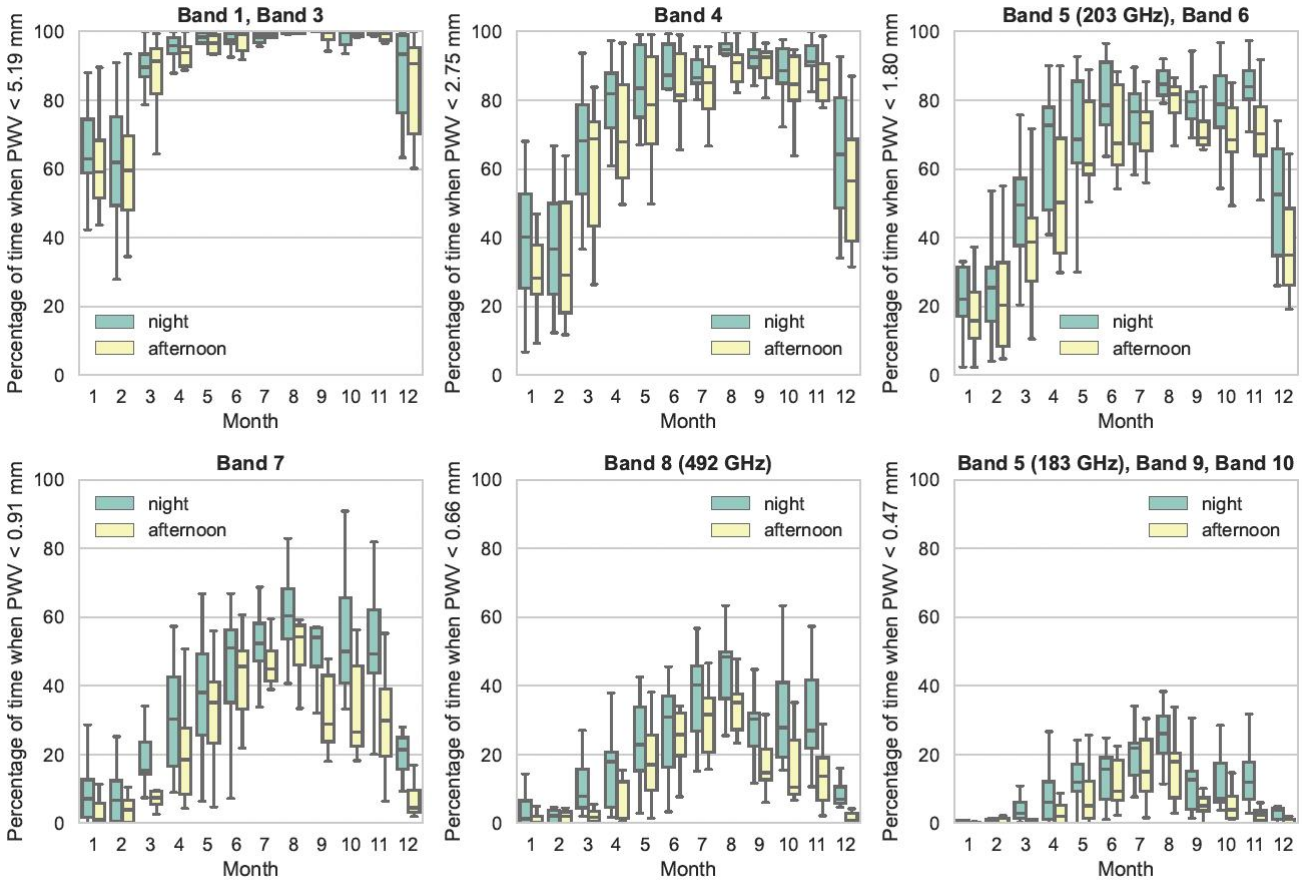

Observing Strategy: Array Configurations

Start date	Configuration	Max Baseline
1 Oct 2026	C-3	0.50 km
20 Oct 2026	C-2	0.31 km
10 Nov 2026	C-1	0.16 km
30 Nov 2026	C-2	0.31 km
20 Dec 2026	C-3	0.50 km
10 Jan 2027	C-4	0.78 km
1 Feb 2027	Maintenance	Maintenance
1 Mar 2027	C-4	0.78 km
20 Mar 2027	C-5	1.4 km
20 Apr 2027	C-6	2.5 km
20 May 2027	C-7	3.6 km
20 Jun 2027	C-8	8.5 km
11 Jul 2027	C-9	13.9 km
30 Jul 2027	C-10	16.2 km
20 Aug 2027	C-9	13.9 km
10 Sep 2027	C-8	8.5 km

ALMA antenna transporter

Credit: NRAO

One day to move an antenna

- Configurations: **C-1 to C-10**
- Maximum baselines: from **0.16 km** to **16 km**

https://almascience.nrao.edu/observing/observing-configuration-schedule/long-term-configuration-schedule

- **Highly encouraged projects**
  - Long baselines (C-7 to C-10) at lower frequencies (Bands 1–4)
  - High-frequency (Bands 8–10)
  - Stand-alone ACA (= no 12-m)

https://almascience.nrao.edu/proposing/proposers-guide